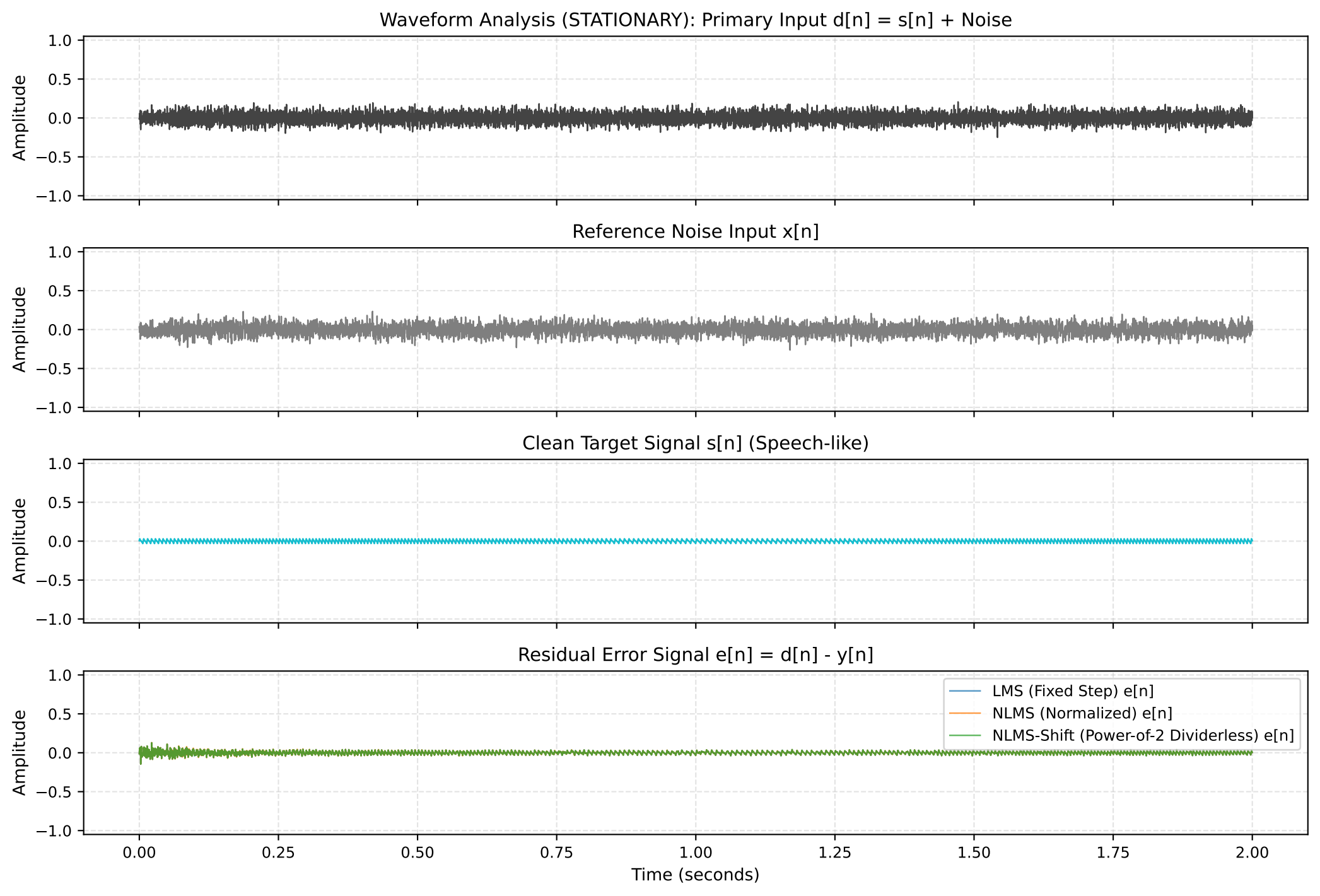

Waveform Analysis (STATIONARY): Primary Input d[n] = s[n] + Noise
Reference Noise Input x[n]
Clean Target Signal s[n] (Speech-like)
Residual Error Signal e[n] = d[n] - y[n]
Amplitude
1.0
0.5
0.0
−0.5
−1.0
LMS (Fixed Step) e[n]
NLMS (Normalized) e[n]
NLMS-Shift (Power-of-2 Dividerless) e[n]
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Time (seconds)